

FDS Lab Exercises

1) Create a Pandas Dataframe using the following dictionary:

```
data = {  
    'Name': ['Alice', 'Bob', 'Charlie'],  
    'Age': [25, 30, 35],  
    'City': ['New York', 'Los Angeles', 'Chicago']  
}
```

Code:

```
import pandas as pd
```

```
Data = {
```

```
    'Name': ['Alice', 'Bob', 'Charlie'],
```

```
    'Age': [25, 30, 35], 'City': ['New York', 'Los Angeles',  
    'Chicago']
```

```
}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```

2) Read a csv file named data.csv into a dataframe and print first 5 rows:

Code:

```
import pandas as pd
```

```
df = pd.read_csv('data.csv')
```

```
print(df.head(5))
```

- 3) Filter rows based on condition from a dataframe df. Print the rows where age > 25.

Code:

```
import pandas as pd
filtered_df = df[df['Age'] > 25]
print(filtered_df)
```

- 4) Add a new column salary with values [50000, 60000, 70000]

Code:

```
import pandas as pd
df['Salary'] = [50000, 60000, 70000]
print(df)
```

- 5) Given a dataframe with columns Department and Salary. Find the average salary per department.

Code:

```
import pandas as pd
```

```
Date = {
```

```
    'Department': ['HR', 'IT', 'HR', 'IT', 'HR'],
```

```
    'Salary': [40000, 50000, 42000, 62000, 50000]
```

```
}
```

```
df = pd.DataFrame(Date)
```

```
Mean_Salary = df.groupby('Department')['Salary'].mean()
```

```
print(Mean_Salary)
```

- b) ~~Replace~~ Replace all occurrences of 'New York' in city with 'NYC'.

import pandas as pd

```
df['city'] = df['city'].replace(New 'New York', 'NYC')  
print(df)
```

- 7) Drop the age column from dataframe.

code:

import pandas as pd

```
df = df.drop(columns=['Age'])
```

```
print(df)
```

- 8) Sort the dataframe by salary in desc order.

Code:

import pandas as pd

```
df_sorted = df.sort_values(by='Salary', ascending=False)
```

```
print(df_sorted)
```

- 9) Check for missing values in the Dataframe.

Code:

import pandas as pd

```
print(df.isnull())
```

```
print(df.isnull().sum())
```

10) Save the dataframe to csv file named Output.csv

Code:

```
import pandas as pd  
df.to_csv  
df.to_csv('Output.csv', index = False)
```

Basic Plotting Functions

1. `Plt. Plot (x, y)` → line plot
2. `Plt. Scatter (x, y)` → Scatter plot (Points)
3. `Plt. Bar (x, height)` → Vertical bar chart
4. `Plt. Barh (y, width)` → Horizontal bar chart
5. `Plt.hist (data, bins = .....)` → histogram
6. `Plt. boxplot (data)` → Box and whisker plot
7. `Plt. pie (data)` → Pie chart

1) The following table shows the number by Date
Science Job Posting recorded over year 2010 to
2020.

Year	Job Postings
2010	150
2011	300
2012	450
2013	600
2014	800
2015	1200
2016	1600
2017	2100
2018	2700
2019	3400
2020	4100

- (i) Create a dataframe for given data
- (ii) Plot line graph showing the trend of Data Science Job Posting.
- (iii) Add suitable title and axis length to graph.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

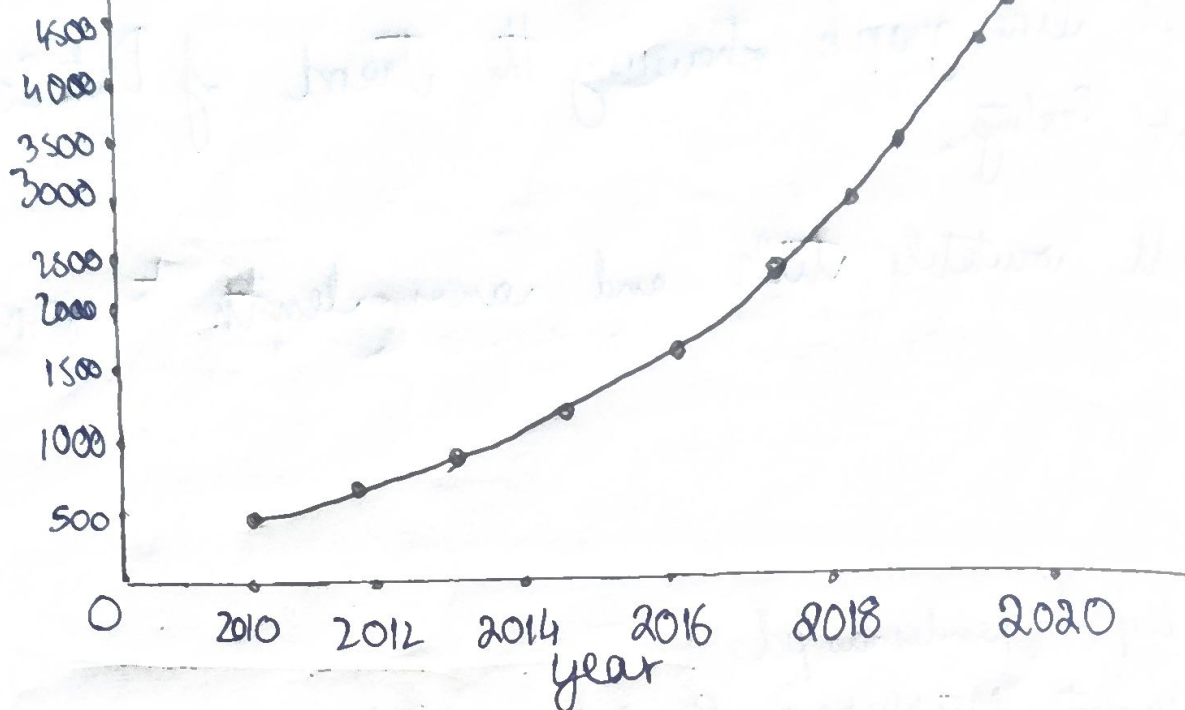
data = {
    'year': list(range(2010, 2021)),
    'Job Posting': [130, 300, 450, 600, 800, 1200, 1600,
                    2100, 2700, 3400, 4200]
}

df = pd.DataFrame(data)
plt.plot(df['year'], df['Job Posting'], marker='o')
plt.title('Trend of Data Science')
plt.xlabel('year')
plt.ylabel('Number of Job Posting')
plt.show()
```

TREND OF DATA SCIENCE

JOB POSTING

No. of job posting

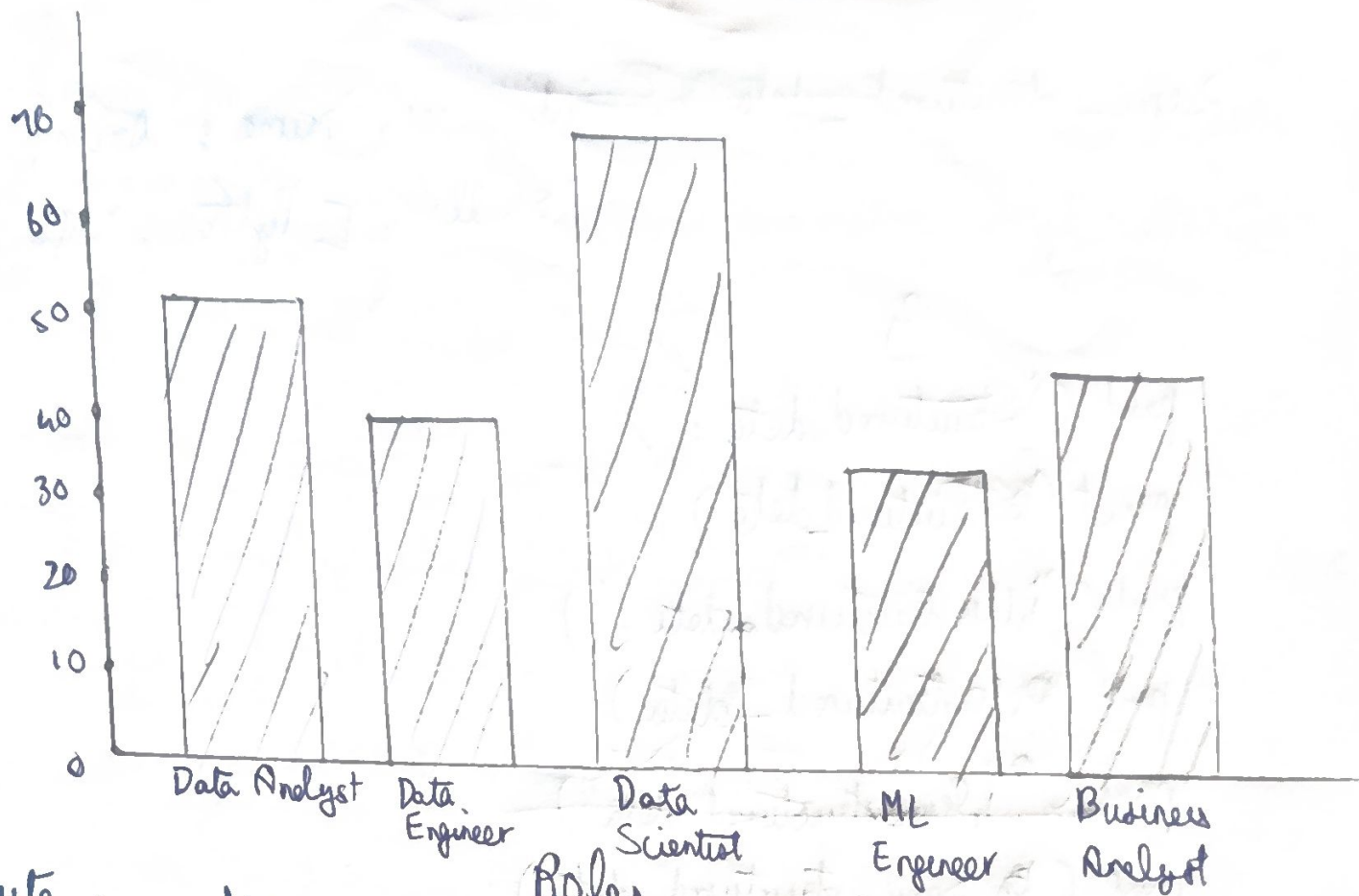


2) Write a python program using Matplotlib to create bar chart that shows distribution of different data science roles with respective counts. Add title to chart.

Code:

```
import matplotlib.pyplot as plt
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer']
counts = [50, 40, 70, 35, 45]
plt.bar(roles, counts)
plt.title('Distribution of Data Science Roles')
plt.xlabel('Roles')
plt.ylabel('Number of employee')
plt.show()
```

No. of Employees



3) Write a python program to demonstrate three main types of Data Structured, Unstructured, Semistructured. Use a Pandas Dataframe for structured data. Print each type of data with clear labels.

Code:

```
import pandas as pd
```

```
Structured - data = pd.DataFrame ( { 'Name': ['Arun', 'Priya', 'Rehul'],
```

```
'Age': [20, 21, 19], 'Department': ['CSE', 'ECE', 'IT']
```

```
})
```

```
Unstructured - data = "Data Science is used to Analyse and understand complex data"
```

```
semi_structured_data = { 'id': 101, 'name': 'Arjun',  
                          'skills': ['Python', 'SQL', 'ML'] }
```

3

```
print ("Structured data :")
```

```
print (Structured_data)
```

```
print ("Unstructured data :")
```

```
print (Unstructured_data)
```

```
print ("Semi-structured data")
```

```
print (Semi_structured_data)
```

Structured_data :

	Name	Age	Department
0	Arjun	20	CSE
1	Priya	21	ECE
2	Rahul	19	IT

Unstructured_data :

Data Science is used to analyse and understand complex data."

Semi-Structured_data :

```
{ 'id': 101, 'name': 'Arjun', 'skills': 'Python' }
```

- 4) Write a python program using cryptography fernet module to demonstrate Symmetric Key Encryption and Decryption. Print original, encrypted, decrypted data.

Code:

```
from Cryptography.fernet import fernet
key = fernet.generate_key
f = fernet(key)
token = f.encrypt(b"Rajalakshmi engineering college")
f.decrypt(token)
token
key = fernet.generate_key()
cipher_suite = fernet(key)
plain_text = cipher_suite.encrypt(Plain_text)
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", Plain_text)
print("Encrypted Data:", Cipher_text)
print("Decrypted Data:", Decrypted_text)
```

Original Data : b' Rajalakshmi Engineering College'

Encrypted Data : b'gAAAAAB9eALSvLT09TLE67gt6iq
406vN8X'

Decrypted Data : b' Rajalakshmi Engineering College'

4). ~~Lab~~

Lab

Sales	ID	Height Height	Weight Weight	Age Age	Income	Exp	Dept	Rating	City
42000	1	185	55	22	28000	1	IT	3.2	Ch
48000	2	170	62	25	32000	2	HR	3.8	Ch
55000	3	175	70	28	38000	4	IT	4.1	Md
62000	4	168	65	NA	41000	5	Sales	4.5	Ch
68000	5	NA	78	35	45000	7	IT	4.0	Cb
75000	6	180	NA	42	52000	10	Sales	4.7	Md
77000	7	172	82	48	NA	12	HR	3.5	Ch
82000	8	175	85	52	68000	15	IT	NA	Cb
57000	9	169	68	29	39000	5	IT	3.9	Md
NA	10	171	72	31	43000	6	Sales	4.2	Ch
76000	11	178	90	67	72000	20	HR	2.8	Cb
91000	12	182	105	75	85000	25	Sales	2.5	Md